⚡ CyberTrader – Run Guide

1️⃣ Start Backend (RabbitMQ + workers)

Open **Conda terminal**, activate your environment, then:

```
python main.py
```

- Connects to RabbitMQ
- Starts price fetchers (`fetch_prices`, `realtime_data`)
- Starts strategies (`multi_ticker_runner`, `reversal_sniper`, `bottom_signal_detector`)
- Publishes signals into RabbitMQ

2️⃣ Start GUI (Frontend)

Open another terminal (same env), then:

```
python CyberNinjaTrader.py
```

- Connects to RabbitMQ
- Opens GUI (ReversalRadarTab)
- Displays signals: **BOTTOM, REVERSAL, BREAKOUT, MOMENTUM, THRESHOLD, UNKNOWN**
- Lets you toggle strategies with checkboxes
- Updates `shared/watchlist.json` if you add/remove tickers

3️⃣ Monitor Logs

- `signal_logger.py` automatically writes every signal to logs (and optionally file).
- `shared/watchlist.json` updates whenever GUI changes tickers.

- `shared/config.json` holds system settings (RabbitMQ, API keys, thresholds).

🛑 Stop System

- Close GUI window (`CyberNinjaTrader.py`).
- In backend terminal (`main.py`), press:

```
CTRL + C
```

to stop RabbitMQ workers.

🖌️ Notes

- **Legacy code** → safely archived in `/legacy/` (not deleted).
- **Single source of truth** →
  - `shared/config.json` = system settings
  - `shared/watchlist.json` = tickers + metadata
- **No txt files left** → everything uses JSON.

👉 With this setup:

- Backend = brains 🧠
- RabbitMQ = nervous system ⚡
- GUI = eyes 👀

🛠️ CyberTrader – Failure Recovery Checklist

🚨 If GUI (CyberNinjaTrader.py) crashes:

1. Close any frozen windows.
2. Re-run:

```
python CyberNinjaTrader.py
```

3. ✅ Signals will repopulate from RabbitMQ (you don't need to restart the backend).

🚨 If Backend (main.py) crashes:

1. Press **CTRL + C** if it's frozen or stuck.
2. Re-start backend:

```
python main.py
```

3. Then re-start GUI in a separate terminal:

```
python CyberNinjaTrader.py
```

4. ✅ RabbitMQ queues will refill, signals flow resumes.

🚨 If RabbitMQ service itself crashes:

1. Stop all running parts (`main.py` + GUI).
2. Restart RabbitMQ (depends on install):
    a. Docker:

```
docker restart rabbitmq
```

b. Local service:

```
sudo systemctl restart rabbitmq-server
```

3. Once RabbitMQ is live again:
   a. Start backend: `python main.py`
   b. Start GUI: `python CyberNinjaTrader.py`

🚨 If Config / Watchlist files break (corrupted JSON):

1. Go to `shared/`.
2. Restore backup:
   a. Claude built **auto JSON backups** during refactor. Look for `watchlist_backup_*.json` or similar.
3. Replace broken file with last known good backup.
4. ✅ Restart system.

🚨 If You're not sure what broke:

1. Check logs in `signal_logger.py` (it will usually show the error).
2. Open `shared/config.json` and `shared/watchlist.json` to confirm they're still valid JSON.
3. Restart whole system in order:
   a. RabbitMQ
   b. Backend (`main.py`)
   c. GUI (`CyberNinjaTrader.py`)

👉 With this, you'll never be "stuck." The system is modular — each part can restart independently.